

Question	Marking Guidance	Mark	Comments
07.1	1. Less (oxygen) loaded at high pO_2 (compared with no CO) OR Maximum Hb saturation is 50% (oxygen); 2. (At low pO_2) Hb has higher affinity (for oxygen); 3. Hb has more oxygen at low pO_2 OR Hb unloads less oxygen at low pO_2;	3 (1 x AO2, 2 x AO3)	Ignore references to 'binding sites occupied by CO' 1. Accept 'in lungs' for 'high pO_2' 1. Accept 'levels (off)' OR 'plateau at 50%' for maximum 1 Accept 'only 50%' for 'maximum (of) 50%' 3. Accept 'in (respiring) cells/tissues' for 'low pO_2/cells' 3. Ignore 'more loading of oxygen at low pO_2' 3. Accept 'less readily' for less

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07.2	(For scientists' suggestion) 1. Children exposed (to CO) for 8 hours; (Against scientists' suggestion) 2. Children exposed (to CO) for 1 hour OR Adults exposed (to CO) for 1 hour and 8 hours; (But) 3. Maths model may not be accurate OR Maths model did not use people; 4. (Recommendation for people) only at rest OR Unknown effects of exercise (on CO uptake); 5. (People vary in) size/age/ethnicity OR Different sex; 6. No statistical test to show if differences are significant; 7. Unknown effects on CO exposure for people with respiratory disease OR Unknown effects on CO exposure for smokers; 8. Might not be able to reduce CO (concentration) in air below 10(mg m⁻³);	3 max (3 x AO3)	Award 3 max only if answer contains MP1 OR MP2 3. Accept '(based on) invalid assumptions' for 'may not be accurate' 3. Accept test OR experiment OR investigation for 'maths model' 5. Accept gender 6. Accept no SDs to show if differences are significant 7. Accept any named respiratory disorder; eg asthma OR coronary heart disease
